

6G Weekly Survey — May 18 - May 24, 2026

16 items across 6 topics, ranked by importance.

AI-native air interface

ChannelAgent-Empowered Electromagnetic Space World Model: A Case Study on Agent-Driven Channel Generation for 6G AI-Native Air Interface

Mingyue Li, Li Yu, Yuxiang Zhang, Heng Wang, et al. · arXiv eess.SP · 2026-05-14

★★★★■ Introduces a genuinely novel paradigm (agent-based world models) for AI-native air interfaces with credible execution on channel generation, though broader validation across diverse 6G tasks is needed.

This paper proposes a ChannelAgent-empowered electromagnetic space world model for 6G, organizing wireless intelligence as a closed loop of sensing, agent-based reasoning, and execution with feedback. As a case study, they instantiate agent-driven channel generation through path loss prediction using a task-oriented feature selection mechanism that combines reinforcement learning policy adaptation with evolutionary search. Simulations demonstrate superior performance in single and multi-scenario tasks compared to conventional approaches.

Contribution: First application of an agent-based architecture with task-oriented feature selection (RL + evolutionary search) to autonomous channel modeling, moving beyond static AI models toward closed-loop adaptive wireless intelligence.

Relevance to your work: Directly relevant to AI-native air interface design; the agent-driven, task-adaptive framework could extend to CSI compression, beam management, or learned PHY where scenario heterogeneity and autonomous adaptation are critical.

Model Forensics in AI-Native Wireless Networks: Taxonomy, Applications, and Case Study

Pengyu Chen, Weiyang Li, Jin Xu, Jiacheng Wang, et al. · arXiv eess.SP · 2026-05-14

★★★★■ Addresses a critical but under-explored security gap as AI models become integral to 6G physical layer and network functions; provides actionable framework with demonstrated techniques that credible research groups and 3GPP security working groups will need to consider.

This paper presents a taxonomy and framework for model forensics in AI-native wireless networks, addressing security risks from model tampering, malicious functions, and provenance verification. The authors categorize forensic techniques into authenticity verification, malicious function identification, and accountability tracing, then demonstrate concrete workflows using RF fingerprinting with watermark authentication and backdoor detection. Experimental results show these techniques can support anomaly assessment and trustworthy operation of AI models deployed in wireless systems.

Contribution: First systematic taxonomy and application framework for model forensics specifically targeting AI-native wireless networks, demonstrating practical watermarking and backdoor detection workflows for RF fingerprinting models.

Relevance to your work: Directly relevant to your AI-native air interface work—as you deploy neural receivers and learned PHY components, this forensics framework becomes essential for ensuring model integrity, detecting adversarial tampering, and establishing trust in production AI-RAN deployments.

ISAC (sensing + communication)

Multimodal Learning for MIMO Beam Prediction Based on Variational Inference

Zijian Zheng, Wenqiang Yi, Hyundong Shin, Arumugam Nallanathan · arXiv eess.SP · 2026-05-14

★★★★■ Addresses a critical deployment barrier (labeled multimodal data scarcity) for ISAC beam management with a principled probabilistic framework and strong empirical validation on a recognized dataset.

The paper addresses beam prediction for massive MIMO in ISAC systems by proposing a variational-inference framework that decouples feature extraction from cross-modal semantic alignment. A two-stage training strategy first learns representations from abundant unimodal data (e.g., single sensor types), then performs alignment on limited multimodal samples. On DeepSense6G, the method achieves competitive accuracy using only 20% of the multimodal training data compared to end-to-end baselines.

Contribution: The variational-inference decoupling enables practical multimodal beam prediction under severe multimodal data scarcity, achieving 5x data efficiency over conventional joint training.

Relevance to your work: Directly relevant to your ISAC and beam management research; the data-efficient multimodal learning strategy could inform designs for sensor-aided CSI acquisition in beyond-5G systems where paired training data is expensive.

SimART: A Unified and Open Real-world Multimodal Simulation Platform for 6G Integrated Sensing and Communication

Kang Yan, Yuqi Cao, Jiaqi Li, Luping Xiang, et al. · arXiv eess.SP · 2026-05-13

★★★★■ Solves a genuine infrastructure gap in ISAC prototyping, enables reproducible multimodal dataset generation, and the open release could accelerate community progress on sensing-communication co-design. SimART addresses the fragmented tooling landscape in 6G ISAC research by integrating robotics simulators, ray-tracing engines, and wireless evaluation tools into a unified pipeline orchestrated via ROS. The platform synchronizes multimodal streams (vehicle dynamics, onboard sensors, site-specific channels) through a shared clock and coordinate frame, recording everything into reproducible rosbag files. A case study on vision-aided beam prediction demonstrates cross-modal fusion, and the system includes automated scene construction from OpenStreetMap plus channel knowledge map generation.

Contribution: First open, modular simulation framework that co-simulates vehicle mobility, onboard perception, and site-specific wireless propagation in a single reproducible workflow for ISAC research.

Relevance to your work: Directly relevant: you need realistic ISAC datasets for training AI-native air interfaces, and SimART's channel-knowledge-map abstraction naturally feeds into learned beam management and joint sensing-communication PHY design.

Antenna Tilt Failure Detection and Estimation via Integrated Sensing and Communications

Samed Kesir, Batuhan Kaplan, Emre Arslan, Ahmet Faruk Coskun · arXiv eess.SP · 2026-05-14

★★★★■ Practical application of ISAC principles to a real operational problem (network maintenance), though incremental rather than advancing core ISAC sensing or communication theory.

This paper proposes using ISAC principles to detect and estimate antenna tilt failures in narrow-beam 5G systems by monitoring gain shifts in environmental clutter returns, eliminating the need for external tilt sensors. The method leverages static clutter as geometric anchors and operates with standard 5G NR frames across different waveforms. Simulations demonstrate feasibility for autonomous network self-healing.

Contribution: First sensor-free antenna tilt monitoring framework exploiting ISAC radar functionality and environmental clutter structure for autonomous fault detection in narrow-beam systems.

Relevance to your work: Directly relevant as a concrete ISAC use case demonstrating how integrated sensing functionality can solve operational challenges in massive MIMO and narrow-beam 5G/6G deployments beyond traditional communication or standalone sensing objectives.

Channel modeling / GAN-based

Electromagnetic Signal and Information Theory: A Continuous-Aperture Array Perspective

Zhaolin Wang, Chongjun Ouyang, Kuranage Roche Rayan Ranasinghe, Shuai S. A. Yuan, et al. · arXiv cs.IT · 2026-05-13

★★★★■ This work establishes foundational analytical frameworks for holographic MIMO and extremely large-scale antenna systems critical to 6G sub-THz communications, though it is primarily pedagogical rather than presenting breakthrough algorithmic or performance results.

This tutorial reviews continuous-aperture arrays (CAPAs) for wireless systems evolving toward larger, denser, higher-frequency antennas, modeling them as spatially continuous electromagnetic fields rather than discrete antenna elements. The paper covers electromagnetic foundations, channel modeling (line-of-sight and multipath), beamforming, channel estimation, and capacity analysis, emphasizing the transition from finite-dimensional discrete arrays to infinite-dimensional continuous-aperture formulations governed by Maxwell's equations. It presents tools like wavenumber-domain methods, functional analysis, and compressive sensing to make these infinite-dimensional problems tractable while preserving physical accuracy.

Contribution: First comprehensive tutorial unifying electromagnetic signal and information theory for continuous-aperture arrays, providing physics-consistent analytical frameworks that bridge Maxwell's equations with information-theoretic limits and practical signal processing for extremely large antenna systems.

Relevance to your work: Directly relevant to your Beyond-5G physical layer research: CAPAs underpin sub-THz waveform design and massive MIMO evolution toward holographic surfaces, and the continuous-field channel modeling perspective could inform GAN-based generative models that respect electromagnetic physics rather than purely statistical correlation structures.

6G Physical Layer

Joint Phase Noise and Channel Estimation for OTFS

Stephen McWade, Arman Farhang · arXiv eess.SP · 2026-05-14

★★★★■ OTFS is a leading 6G waveform candidate, and this rigorously addresses a practical impairment (phase noise) with oscillator-specific modeling and a tractable joint estimation solution.

This paper analyzes phase noise effects in OTFS modulation, showing it causes severe inter-Doppler interference (IDI) that common-phase-error compensation cannot address. The authors derive SINR expressions for three oscillator types (free-running, continuous-time PLL, discrete-time PLL) and propose a Wiener filtering approach for joint channel and phase noise estimation. Simulations demonstrate up to 8 dB BER improvement over existing methods that assume known single-tap channels.

Contribution: First joint estimator for multi-tap channels and phase noise in OTFS that explicitly addresses inter-Doppler interference rather than just common-phase error.

Relevance to your work: Directly relevant to your work on 6G waveforms and AI-native PHY—this identifies a fundamental OTFS challenge that learned receivers or neural joint estimators could potentially address more effectively.

Holographic Airy Beamforming: Curved Trajectory Optimization for Blockage-Resilient Terahertz Communications

Xinyuan Hu, Boya Di, Lingyang Song · arXiv eess.SP · 2026-05-15

★★★★■ Addresses critical THz blockage problem for 6G with novel holographic surface approach and demonstrates significant performance gains, though practical RHS implementation challenges remain unaddressed.

This paper addresses terahertz near-field blockage by using reconfigurable holographic surfaces (RHS) to generate Airy beams that propagate along curved parabolic trajectories around obstacles. The key innovation is exploiting RHS's sub-wavelength element spacing to enable adjustable effective aperture, allowing parabolic offsets within the antenna aperture for flexible trajectory design. Geometry-based trajectory optimization demonstrates over 10 dB received power improvement for blocked users compared to conventional phase-controlled arrays.

Contribution: First application of RHS amplitude-controlled holographic beamforming to generate Airy beams for THz blockage circumvention, exploiting adjustable effective aperture for enhanced trajectory design freedom unavailable in traditional arrays.

Relevance to your work: Directly relevant to your 6G physical layer interests—combines advanced antenna architectures (holographic surfaces) with sub-THz propagation challenges and offers a fresh beamforming paradigm beyond conventional massive MIMO techniques.

Map2APS: A Physically Grounded Benchmark for Direct Angle Power Spectrum Prediction from Urban Geometry

Junxi Huang, Xiucheng Wang, Nan Cheng, Kailong Wang, et al. · arXiv eess.SP · 2026-05-14

★★★★■ Provides a rigorously constructed public dataset with ground-truth propagation physics that enables reproducible research on environment-aware beam management and MIMO, a capability critical for AI-native air interfaces in 6G.

Map2APS introduces a large-scale benchmark dataset containing 2.55 million transmitter-receiver samples across 51 urban maps, generated via intelligent ray-tracing to predict angle power spectra (APS) directly from map geometry. The dataset enforces strict cross-map splits to test generalization to unseen urban layouts, and includes a baseline model (MS-AReg) achieving 0.948 cosine similarity and 1.20° peak location error at 0.101 ms inference latency. Evaluation metrics include both spectral fidelity and task-relevant dominant-direction accuracy for beam management.

Contribution: First standardized large-scale benchmark for direct map-to-APS prediction with physically grounded ray-tracing data and cross-map generalization evaluation, addressing a gap in environment-aware learning for directional channel characterization.

Relevance to your work: Directly supports your work on AI/ML for the physical layer by providing a benchmark for learning-based beam management and channel-geometry mapping, essential for massive MIMO and integrated sensing-communication applications.

A Computationally Efficient Reciprocal Effective Roughness Model for Diffuse Scattering

Giacomo Melloni, Enrico M. Vitucci, Vittorio Degli Esposti, Samuel Berweger, et al. · arXiv eess.SP · 2026-05-18

★★★★■ Solid contribution addressing computational tractability of ray-tracing for 6G frequencies where diffuse scattering is significant, though focused on modeling infrastructure rather than breakthrough channel characterization or system design.

This paper presents a computationally efficient diffuse scattering model for ray-tracing in mmWave and sub-THz channels, where surface roughness becomes comparable to wavelength and diffuse scattering can contribute up to 40% of received power. The proposed reciprocal Effective Roughness model maintains physical consistency while achieving an order-of-magnitude speedup over existing methods. Validation across eight materials shows equal or slightly improved accuracy compared to conventional approaches.

Contribution: An order-of-magnitude faster directive and reciprocal diffuse scattering model that preserves the physical structure of Effective Roughness while enabling scalable ray-tracing for mmWave/sub-THz frequencies.

Relevance to your work: Directly relevant to sub-THz channel modeling for 6G, providing computationally tractable tools for generating realistic training data and simulations that could support your work on AI-native physical layers and learned channel models.

Fast 5G Signal Acquisition by Using Non-Uniform Sampling

Alejandro Gonzalez Garrido, Carla Amatetti · arXiv eess.SP · 2026-05-18

★★★★■ Solid incremental work demonstrating practical complexity-performance trade-offs for 5G synchronization, but limited to known sequences and does not address learning-based approaches or extend beyond initial acquisition.

This paper frames 5G signal acquisition as a parametric inference problem in delay-Doppler space and proposes multi-coset non-uniform sampling to reduce complexity. Instead of reconstructing the full Nyquist-rate signal, the method operates directly on compressed samples using a generalized likelihood ratio test with optimized coset patterns. Evaluations on 5G NR PSS/SSS synchronization demonstrate 2.8x to 34.2x reductions in mean acquisition time relative to uniform sampling, with quantified estimation accuracy

trade-offs.

Contribution: The key novelty is reframing synchronization as compressed-domain inference rather than waveform reconstruction, enabling direct operation on multi-coset samples with explicit coset pattern optimization for delay-Doppler acquisition metrics.

Relevance to your work: This connects to your work on AI-native air interfaces by addressing sub-Nyquist acquisition, which could inform neural receiver designs that jointly optimize sampling and detection, especially for beam management and channel estimation under rate constraints.

TFZ-Tree: An Ultra-Lightweight Waveform Classification Framework for Resource-Constrained Devices

Hao Wang, Kuang Zhang, Yonggang Chi, Tianqi Zhao, et al. · arXiv eess.SP · 2026-05-15

★★★★■ Addresses a practical deployment gap for multi-waveform 6G IoT receivers with a novel lightweight approach, but scope is narrow (IoT waveforms) and multipath performance degrades notably.

This work addresses waveform-type classification (OFDM, OTFS, LoRa, etc.) for 6G IoT receivers, proposing an ultra-lightweight framework using time-domain features and a Z-test-based decision tree (ZTree) instead of deep neural networks. Tested on ten candidate waveforms, it achieves 99.5% accuracy under AWGN and 87.4% under multipath with <4 ms inference latency on x86, targeting deployment on resource-constrained MCUs. The approach fills a gap between heavy DNN-based methods and the practical need for real-time waveform identification before demodulation.

Contribution: First real-time classifier for ten IoT waveform types using hypothesis-test-driven tree splitting with low-complexity features, avoiding deep learning overhead for resource-limited terminals.

Relevance to your work: Relevant to your AI-native PHY work as a counterpoint: demonstrates when classical statistical methods can outperform neural approaches for waveform identification under strict resource constraints, though your focus on learned end-to-end systems may require hybrid approaches.

Semantic communication

Not All Symbols Are Equal: Importance-Aware Constellation Design for Semantic Communication

Albert Shaju, Christo Kurisummootil Thomas, Mayukh Roy Chowdhury · arXiv eess.SP · 2026-05-14

★★★★■ Well-motivated extension of semantic communication to physical layer design with rigorous analysis and strong empirical gains, though evaluation is limited to classification tasks on standard vision/audio datasets rather than wireless-specific scenarios.

The paper proposes a joint semantic-physical layer framework that assigns M-QAM constellation points based on semantic importance rather than uniform Gray coding. A VQ-VAE extracts discrete latent concepts, a semantic criticality indicator scores them by task relevance, and a reinforcement learning agent selects transmission subsets under channel constraints. The approach achieves near 100% semantic protection probability for critical symbols versus 50% for standard constellations at equivalent spectral efficiency.

Contribution: First work to prove Gray-coded constellations are strictly suboptimal for semantic communication and to design learned constellations that protect task-critical symbols by jointly optimizing symbol placement with semantic importance scores and co-occurrence statistics.

Relevance to your work: Directly relevant to your semantic and task-oriented communication research; the joint optimization of physical-layer mapping with task criticality aligns with AI-native air interface design, though applicability to continuous channel estimation or ISAC tasks remains open.

Joint Communication and Computation Design for Mobile Embodied AI Network (MEAN)

Chenliang Wu, Zhouxiang Zhao, Jiaxiang Wang, Ruopeng Xu, et al. · arXiv eess.SP · 2026-05-14

★★★★ Solid application of semantic communication principles to multi-agent systems with practical energy-efficiency focus, though limited novelty in the semantic communication techniques themselves and unclear generalization beyond the specific dual-mode setup.

This paper addresses energy-efficient collaborative task execution for mobile embodied AI agents communicating wirelessly with a base station. Agents can switch between BS-assisted semantic communication for collaborative decisions and local independent operation based on SNR and collaboration benefits. The authors formulate a mixed-integer nonlinear program jointly optimizing semantic compression ratio, transmit power, collaboration scale, and operating mode selection, solving it via closed-form solutions for convex subproblems and greedy enumeration for mode assignment.

Contribution: Joint optimization framework combining semantic compression, power allocation, and mode selection for multi-agent embodied AI systems, with analytical solutions demonstrating energy savings through adaptive collaboration strategies.

Relevance to your work: Directly relevant as an application of task-oriented/semantic communication to coordinated AI agents, demonstrating how semantic compression ratios can be optimized jointly with PHY-layer parameters in multi-agent scenarios.

Toward Communication-Efficient Space Data Centers: Bottlenecks, Architectures, and New Paradigms

Minghao Sun, Zehui Chen, Jinbo Hou, Kezhi Wang, et al. · arXiv cs.NI · 2026-05-12

★★★★ Solid conceptual work connecting semantic communication to an emerging non-terrestrial application (SDCs), though evaluation remains preliminary and deployment is speculative beyond 2030.

This paper examines Space Data Centers (SDCs) that host AI training and inference in orbit to exploit solar power and radiative cooling, but face a fundamental bottleneck: petabit-scale internal data exchange versus gigabit-scale ground-space links. The authors propose semantic communication to transmit compact task-relevant representations instead of raw data, and evaluate a multi-layer heterogeneous SDC architecture with relay satellites and orbital computing nodes under energy and thermal constraints. They demonstrate feasibility and outline open challenges for scalable orbital AI infrastructure deployment.

Contribution: First systematic analysis of communication bottlenecks in orbital data centers and proposal of semantic communication as the enabling paradigm to bridge the petabit-gigabit gap between internal compute fabric and ground uplinks.

Relevance to your work: Directly relevant to your NTN and semantic communication interests—demonstrates a concrete use case where task-oriented compression is not optional but architecturally necessary for orbital AI infrastructure.

NTN / satellite

Impact of Terrestrial Blockage on the Coverage of Integrated Satellite-Terrestrial Networks

Joon-Young Park, Byungju Lim, Young-Chai Ko · arXiv eess.SP · 2026-05-13

★★★★ Solid analytical work addressing a real deployment challenge for 6G NTN integration, though modeling assumptions (Boolean blockage, downlink only) limit immediate applicability to live systems.

This paper develops a stochastic geometry framework with Boolean blockage models to analyze downlink coverage in integrated satellite-terrestrial networks. The analysis reveals that terrestrial blockages attenuate satellite signals but simultaneously suppress aggregate interference, yielding a net coverage benefit. Analytical results demonstrate that satellite-terrestrial integration improves coverage across diverse urban-density scenarios under heterogeneous blockage conditions.

Contribution: First rigorous stochastic geometry treatment quantifying the dual effect of urban blockages on satellite links—both attenuating desired signals and shielding against interference—in integrated NTN-TN deployments.

Relevance to your work: Directly relevant to your NTN/LEO/direct-to-cell interests; the blockage-aware coverage framework informs system design for satellite-terrestrial integration scenarios and complements channel modeling work.

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